

Investigating Grain Size Influence on the Stress-Dependent Permeability of Porous Rocks Using Particulate Discrete Element Method

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ABSTRACT: Reservoir in-situ stresses continuously evolve during the reservoir life cycle because of the production of hydrocarbons, injection of water and CO₂ for enhanced oil recovery, storage of CO₂, and nuclear waste disposal. The induced changes to the stress state deform the reservoir rock, including the pores and fractures in the rock, and hence change the permeability of the rock. Studies demonstrate that grain size significantly affects the mechanical properties and strength of rock, yet little research exists on the effect of grain size on stress-sensitive seepage behavior.

In this study, we investigated the effect of grain size on the stress-dependent flow behavior of a porous rock using the three-dimensional discrete element method. We modeled the porous rock geometry using randomly generated rigid spherical grains of given size distribution enclosed in a cuboid. Particle-particle interactions and wall-particle interactions are governed by the Flat-Joint contact model. Rigid boundary walls are used, and a constant strain rate is applied to the boundaries to reach a pre-defined confining stress state. Subsequently, we used the pore scale finite volume method to model the steady-state fluid flow of single-phase fluid at each confining stress state of the model.

The permeability-stress relationship data is best described by the exponential model, and the exponent of the exponential model (stress-sensitive coefficient) indicates how rapidly the permeability changes with stress. Our results show that the stress-sensitive coefficient varied very little with the grain size and shows a general decreasing trend with an increase in the mean grain size of the spherical pack. We argue that the effect of other parameters such as pore morphology, micro-cracks, and cementation has a greater influence on the stress-sensitive permeability than the grain size alone.

1 INTRODUCTION

The subsurface in-situ stresses exhibit dynamism, which is attributed to the influence of several anthropogenic factors such as reservoir depletion during hydrocarbon extraction, CO₂ sequestration, geothermal activities, nuclear waste disposal, and natural causes arising from tectonic movements. Such stress variations significantly impact the physical properties of rocks, particularly their porosity, permeability, and dynamic elastic properties. Given the practical applications associated with these effects, the geoscience community has shown significant interest in this aspect.

Stress-dependent permeability has been studied through experimental, empirical, and analytical modeling and numerical simulation techniques. The initial evidence of stress-sensitive permeability was provided by experimental studies (Fatt and Davis, 1952), which were followed by studies that aimed to understand the effect of pore pressure and confining pressure independently on stress-sensitive permeability and proposed effective stress laws (Zoback and Byerlee, 1975).

Laboratory experiments investigating the effect of stress-sensitive permeability on reservoir rocks have been con-

ducted on various rock types, including shales (Al Ismail and Zoback, 2016; Saurabh et al., 2022), sandstones and limestones (Wang et al., 2018; Teklu et al., 2018), fractured rocks (Zhou et al., 2019; Baghbanan and Jing, 2008), coals (Cao et al., 2021), and synthetic rocks (Zou et al., 2013).

The results of several experiments have demonstrated that the sensitivity of stress to permeability exhibits a broad range of variations. In some cases, the stress-induced permeability reductions can be as high as 98 percent of the original value (Teklu et al., 2018). Field studies have also been conducted to analyze production behavior, taking into account the effect of stress-sensitive permeability. These studies have revealed that permeability reductions can lead to a decrease of up to 50 percent in the initial stage gas production rate (Davies and Davies, 2001; Raghavan and Chin, 2002). To calculate the stress-induced changes in permeability, various mathematical models have been developed, including analytical expressions based on idealized geometries (Gangi, 1978) and empirical equations (David et al., 1994; Shi and Wang, 1986).

Several previous studies have demonstrated that grain size is a significant factor in controlling the mechanical properties of geological rocks. A number of researchers have conducted experimental investigations to examine the effect of grain size on the mechanical properties of various geological rock types (Hugman III and Friedman, 1979; Přikryl, 2001; Räisänen, 2004). It has been observed that, for crystalline rocks, the rock becomes weaker in terms of its strength as the grain size increases (Přikryl, 2001; Fredrich et al., 1990). Experimental studies on crystalline rocks have led to the development of regression equations that establish the relationship between the mechanical properties of the rock and the grain size of its constituent minerals. One widely accepted correlation is that the UCS of the rock varies linearly with the inverse square root of the mean diameter. However, such a strong correlation is not evident in the case of porous sandstone rocks. The results of experimental studies carried out by Atapour and Mortazavi (2018) on artificial sandstone rocks suggest that the factors that have a larger impact on the UCS of these rocks are the porosity and the proportion of cement rather than the grain size.

Textural characteristics play a crucial role in determining the hydro-mechanical behavior of rocks. These characteristics encompass several elements, including mineral composition, grain size and shape, the spatial distribution and orientation of grains, porosity, the presence of microcracks, and the relative proportion of grains to matrix material (Ersoy and Waller, 1995; Folk, 1966). The main focus of our study is to investigate how grain size affects stress-

sensitive permeability behavior. To the best of our knowledge, very little research is conducted to study the effect of grain size on the stress-dependent permeability of granular rocks. The findings will aid in the advancement of research on the modeling of rock granular materials in the management of hydrocarbon and geothermal reservoirs, monitoring of CO₂ sequestration, and geotechnical engineering.

2 MATERIALS AND METHODS

Using the Particle Flow Code 3D (PFC3D) discrete element method (Itasca Consulting Group, Inc., 2021), a model of mineral grains in granular sandstone rock is constructed, and a uniform mesh is generated for fluid flow calculations based on particle geometry. To assess the rock's permeability, a finite volume fluid-solid coupling algorithm is used by combining the particle model and fluid calculation mesh to simulate fluid flow within the particle assemblage.

2.1. Model Generation

The process of modeling a granular assemblage with the PFC3D typically comprises a series of discrete procedural steps. These steps include the initiation of the computational domain, the generation of boundary walls, the distribution of granular entities (also referred to as "balls" or "spheres") within the domain and their material characterization, the specification of appropriate contact models and associated parameters for both ball-ball and ball-wall interactions, the assignment of initial conditions, and the subsequent alteration of system state through application of appropriate boundary conditions.

Table 1: Description of samples and respective grain size ranges used in the present study

Sample ID	Mean Diameter (mm)	Minimum Diameter d_{min} (mm)	Maximum Diameter d_{max} (mm)	$\frac{d_{max}}{d_{min}}$	Number of Grains (N)
S1	1.50	1.000	2.000	2	93422
S2	2.00	1.333	2.667	2	39479
S3	2.50	1.667	3.333	2	20180
S4	3.00	2.000	4.000	2	11629
S5	3.50	2.333	4.667	2	7305
S6	4.00	2.667	5.333	2	4899
S7	4.50	3.000	6.000	2	3449
S8	5.00	3.333	6.667	2	2484

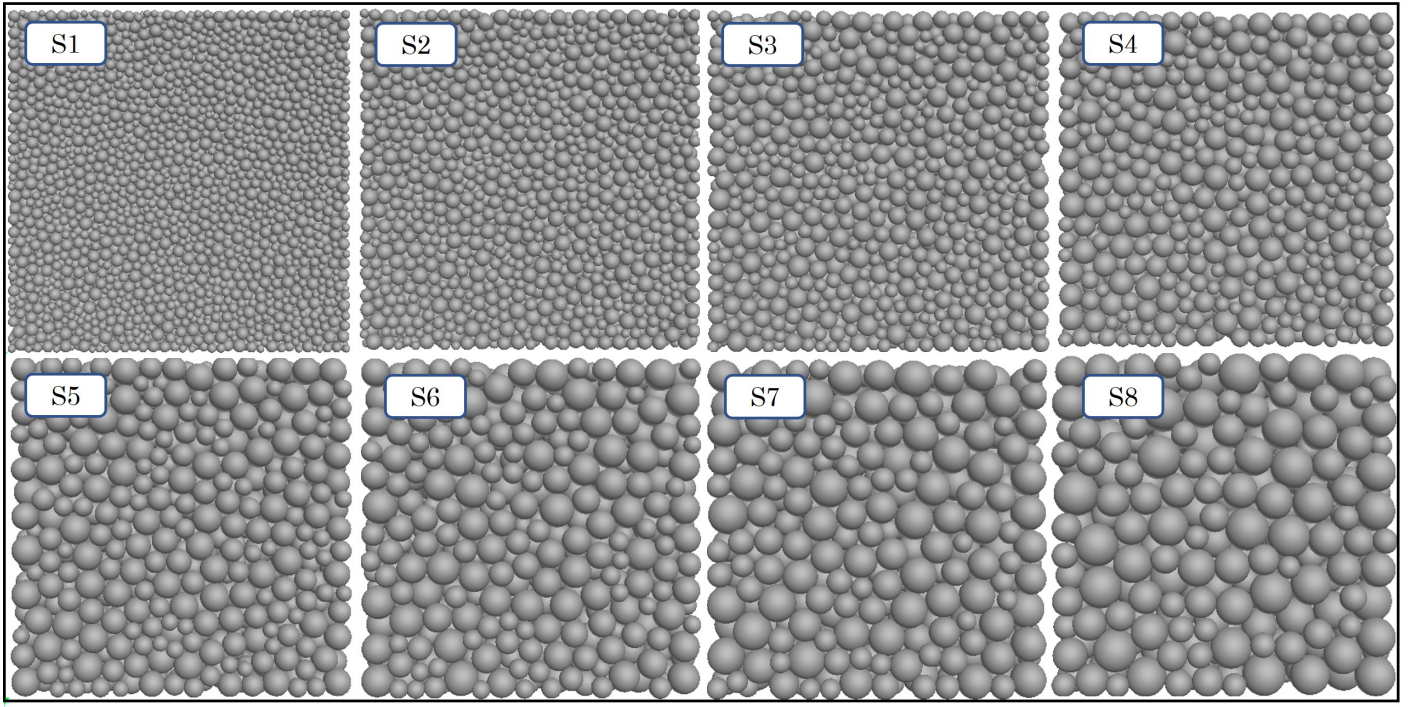


Fig. 1: Arrangement of spheres corresponding to samples with different grain size distributions.

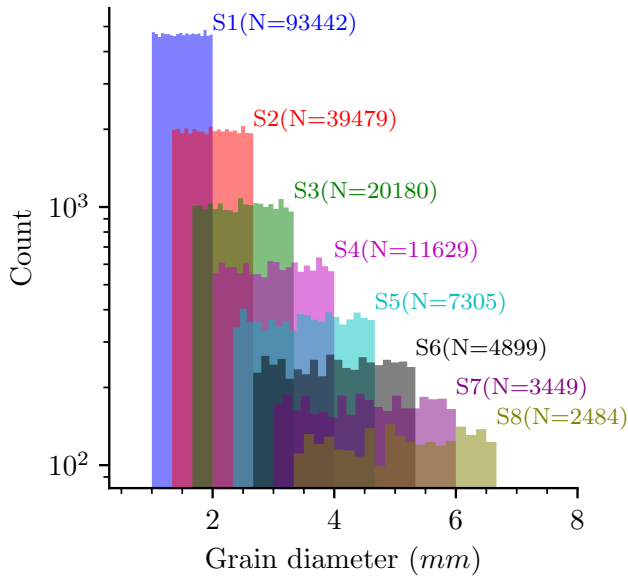


Fig. 2: Grain size distribution of each sample. The total number of grains in each sample is presented inside the brackets.

Initially, a set quantity of spheres is produced in a volume defined by a right square prism enclosed by walls. Following this, a process of force equilibrium is performed to stabilize the system. The number of spheres generated in the given dimensions depends on the target porosity and the ball size distribution. The total number of spheres (N) can be computed using the expression

$$N = \frac{6V(1-\phi)}{\pi d_{avg}^3}; \quad \text{with } d_{avg} = \frac{d_{min} + d_{max}}{2} \quad (1)$$

where ϕ is the target porosity, V is the target volume to be filled, d_{min} , d_{max} and d_{avg} are the minimum, maximum, and average sphere diameters respectively.

The present investigation comprises a sample volume that can be described as a right square prism with dimensions of a height of 101.6 mm and a width of 50.8 mm. Eight distinct grain size distributions are studied, denoted S1, S2, S3, S4, S5, S6, S7, and S8, which exhibit mean grain diameters of 1.5 mm to 5.0 mm with increments of 0.5 mm (figure 1). It is noteworthy that the sample size for each case remains constant, regardless of grain size. Table 1 shows the detailed grain size distributions and the total number of grains generated in each sample. Grain size distribution along with the cumulative distributions are presented in figure 2

Previous research has highlighted the influence of the maximum to minimum grain size ratio (d_{max}/d_{min}), on macro-mechanical properties, such as the UCS, Young's modulus, and Poisson's ratio (Ding et al., 2014; Li et al.,

2018). Notably, it has been observed that these properties tend to decrease as d_{max}/d_{min} increases. Therefore, in the current study, we have chosen to maintain a constant d_{max}/d_{min} ratio of 2 for each sample, along with our desired mean grain sizes, to facilitate an optimal analysis of these macro-mechanical properties (Table 1).

2.1.1 Contact model

Flat-joint contact model developed by Potyondy (2012) is used in the present study. Several researchers used a flat joint contact model to successfully reproduce the mechanical properties of sandstone (Bahaaddini et al., 2021; Potyondy, 2017). In this modeling approach, flat joints represent a theoretical construct that takes the form of circular disks created between the grain contacts when they are separated by closer than the predefined 'flat-joint gap' parameter. These flat joints are further divided into smaller units known as elements, and each element can either be bonded or unbonded. When a flat joint element undergoes deformation, it does so in a linear elastic manner until it reaches its maximum strength limit. A bonded element breaks when the exerted stress is greater than either its tensile strength or shear strength and becomes unbonded. The unbonded elements of the flat joint can undergo further linear elastic deformation accompanied by friction as well as a degree of slip that is determined by Coulomb's criteria. When all the individual elements that make up the interface are considered together, their collective contribution results in an emergent interface behavior that changes over time. This behavior starts with a linear elastic and fully bonded state, where all the elements are completely connected and the interface functions as a single unit. As the elements undergo deformation and stresses, the interface gradually transitions to a partially damaged state, where some of the elements are no longer connected, and the interface starts to function as a partially connected unit. Finally, as the deformation and stress increase, the interface becomes a frictional state. When the relative displacement between the grains becomes larger than the diameter of the flat joint interface, then the contact is broken and removed. A complete list of governing equations and a description of the flat-joint model can be found in Potyondy (2012).

2.1.2 Model calibration

In this study, the micromechanical characteristics of the PFC3D model have been calibrated to reflect the macroscopic properties of a moderate-strength sandstone. The resulting digital rock exhibits uniaxial compressive strength (UCS) values ranging from 55 to 70 MPa, Young's modulus within the range of 13 to 15 GPa, and a Poisson's ratio of approximately 0.21, indicative

of a medium-strength sandstone rock (Deere and Miller, 1966). Table 2 provides the final set of microparameters employed in the investigation.

2.1.3 Fluid flow

The finite volume method is used to simulate the fluid flow in the granular assembly. We used *FiPy* python package that solves finite volume equations (Guyer et al., 2009). Darcy's law can be used to model the movement of fluid through a porous material when the Reynolds number is less than 10. Together with the incompressible fluid condition, the governing equations can be written as (Atangana, 2018)

$$\vec{v} = -\frac{K}{\eta\phi}\vec{\nabla}p \quad (2)$$

and

$$\vec{\nabla} \cdot \vec{v} = 0 \quad (3)$$

Where, $\vec{v}(m/s)$, $K(m^2)$, $\eta(Pa \cdot s)$, $\phi(v/v)$ and $\vec{\nabla}p(Pa/m)$ are the fluid velocity, permeability, dynamic viscosity, porosity, and fluid pressure gradient respectively. These two equations governing the flow of fluids are solved using a set of coarse elements. The velocity of the fluid is approximated as a piece-wise linear function on these elements. This collection of elements, referred to as the fluid mesh, is utilized to determine the porosity by evaluating the extent of overlap between the PFC3D particles and the fluid elements. Each element in the mesh contains a singular value that represents the fluid's distinct properties and state variables, such as velocity and pressure. If necessary, the pressure values at the center of the cell faces can be estimated by interpolating the values calculated at the cell centers. In order to incorporate the effects of the particles on the fluid flow, a permeability value is computed for each element of the fluid mesh based on the porosity of the PFC3D model. The Kozeny-Carman relationship is employed to estimate the value of permeability (Amador and Martin de Juan, 2016)

$$K_e = \frac{1}{180} \frac{\phi_e^3}{(1 - \phi_e)^2} (D_e)^2 \quad (4)$$

where, $K_e(m^2)$, $\phi_e(v/v)$ and $D_e(m)$ are the permeability, porosity, and average grain diameter within the element of the fluid mesh. The one-way coupling has been considered in this study, which means that the change in fluid pressure induces mechanical deformation. By keeping the Reynolds number below 10, the volumetric flow rate has been maintained at a low value of $10^{-6} m^3/s$. This low flow rate results in a fluid pressure drawdown that is always less than 100 Pa, which is an extremely small value as compared to the applied stress. This implies that the mechanical deformation due to fluid flow

is negligible, and therefore the effect of fluid flow on the solid matrix can be ignored.

2.2. Numerical simulation

The study follows a well-defined sequence, as illustrated in the flowchart (Figure 3). First, a model of sphere assemblage is created, and material and contact properties are assigned. This model is then utilized for the UCS test and fluid flow simulations. The UCS test enables the estimation of the mechanical properties and strength of the model. Subsequently, the confining pressure is incrementally increased in steps of 10 MPa. At each confining stress state, the fluid mesh is regenerated due to the boundary position being different at each confining stress state as a result of the strain experienced by the boundaries. It is important to note that the confining pressure is applied to the dry rock without any fluid pressure in the pore spaces. Following each application of confining stress, the sample undergoes volumetric deformation, which causes a reduction in pore volume. This, in turn, has an impact on the fluid flow capacity of the sample.

2.2.1 Measurement of stresses and strains

We measured the axial stress, axial strain, and lateral strain during the simulation during the UCS test. First, the contact forces associated with the top and bottom walls are measured, and then the axial stress is obtained by considering the average of these two forces divided by the cross-sectional area. The axial strain was estimated based on displacement measurements of the top and bottom walls divided by the sample height. The measurement of lateral strain is done using spheres instead of walls. First, the spheres located on the lateral boundaries are grouped. Then the average displacement of each group was utilized to estimate the lateral strain.

3 RESULTS AND DISCUSSION

3.1. Mechanical Properties

Simulation results indicate that mechanical properties are influenced by the size of the grains, displaying a grain-dependent behavior. Specifically, the UCS and Young's modulus exhibit a decreasing trend with increasing grain size, consistent with the general trend observed in crystalline rocks. Figure 6 shows the tensile and shear cracks developed after the UCS test. The findings presented in this figure demonstrate that a significant proportion of the bond breakages observed in the studied specimen are tensile cracks. These cracks appear to coalesce, resulting in the development of fractures, which ultimately lead to the failure of the specimen. Figure 5 depicts stress-strain curves obtained by conducting the UCS test using

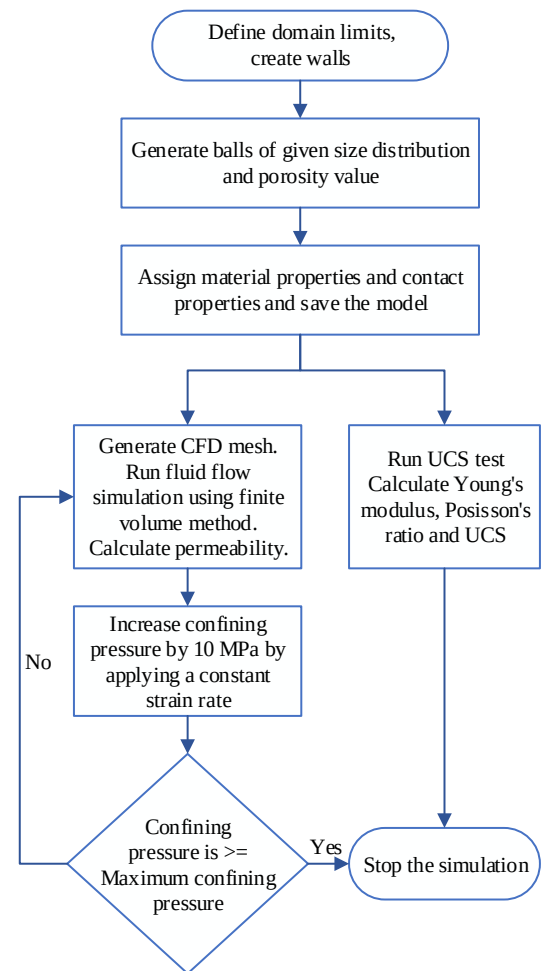


Fig. 3: Workflow used in the present study.

the PFC3D simulation. These stress-strain responses are then utilized to compute the UCS, Young's modulus, and Poisson's ratio. Further, these results are presented in the table 3 and plotted as a function of mean grain diameter in figure 7. It can be seen from figure 7 that UCS is varying from 58 MPa to 69 MPa, and Young's modulus is varying from 13.9 GPa to 15.0 GPa. It is worth noting that the effective modulus of the grain and the normal-to-shear stiffness ratio demonstrate size-independent behavior. Potyondy and Cundall (2004) observed similar behavior of UCS and E decreasing with increasing particle size and they refrain from making definitive claims about the correlation between grain size and mechanical properties. In their numerical simulations, Yu et al. (2018) noted a comparable pattern of UCS reliance on grain size. Their study revealed that grain size leads to various stress concentrations in the rock. When the loading conditions are identical, larger grain sizes produce stronger stress concentrations. This distinct mechanical response can eventually

Ball
Balls (20772)
ball
Element CFD pressure
No objects fit the plot criteria.
CFD Elements (0)
3.0126E+00
2.5000E+00
1.7500E+00
1.0000E+00
2.5000E-01
1.0879E-02
Element CFD Arrow velocity
Maximum: 0.00228734
Scale: 0.0130092
Arrows (1024)
1.0000E-02
8.0000E-03
6.0000E-03
4.0000E-03
2.0000E-03
1.0000E-06

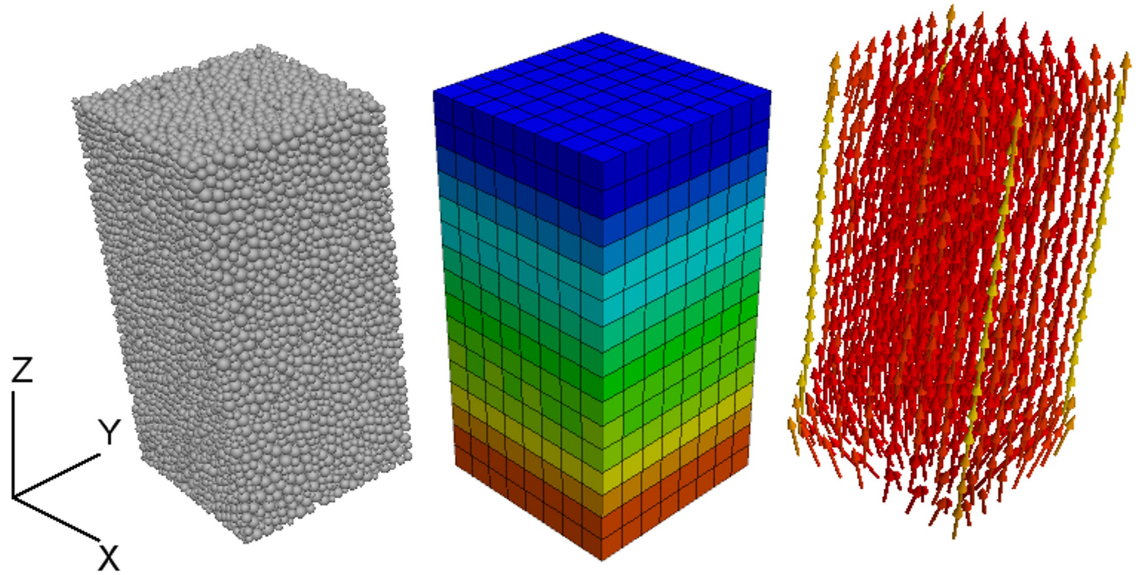


Fig. 4: PFC3D simulation results. Ball arrangement (left), fluid pressure(center) and fluid velocity vectors(right)

Table 2: PFC Parameters

Parameter	Value	Units
Specimen width (X-direction)	50.8	mm
Specimen length (Y-direction)	50.8	mm
Specimen height (Z-direction)	101.6	mm
Ball Density	2650.0	kg/m ³
Ball Damping	0.2	
FJB Effective modulus	15.0	GPa
FJB Normal-to-shear stiffness ratio	4.0	
FJB Tensile strength	20.0	MPa
FJB cohesion	20.0	MPa
FJB Friction angle	15.0	deg
FJB Friction coefficient	0.15	
FJB Surface installation gap	0.2	mm
Fluid density	1000.00	kg/m ³
Fluid dynamic viscosity	0.001	Pa.S

* FJB = Flat Joint Bond Model

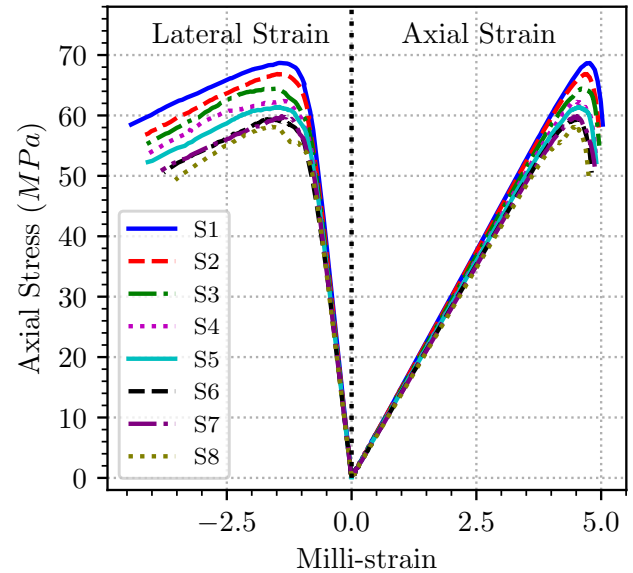


Fig. 5: Stress-strain curves obtained by conducting the UCS test using the PFC3D simulation. Positive values indicate compressive stress and compressive strains.

impact the UCS of the rock.

3.2. Stress dependent permeability

At each confining stress state of the model, fluid flow simulation is conducted as depicted in the flow chart 3. Fluid pressure drawdown, flow rate, and sample dimensions are then extracted using PFC fish functions. These values were subsequently employed to perform a steady-state permeability calculation utilizing Darcy's equation 2. Figure 4 illustrates a sample that contains grains, a fluid pressure field, and fluid velocity vectors. To maintain conciseness, only one fluid flow simulation figure that corresponds to sample S3 is presented here. The permeability evolution with the applied confining stress results is presented in figure 8. It can be seen from the figure that

the permeability of all the samples are decreasing with the increase of confining pressure. In the next section, two popular empirical models are used to fit the permeability results and analyzed the permeability-decreasing behavior.

3.2.1 Empirical Models

In the literature, two commonly used models are the exponential and power-law models. The exponential model, as

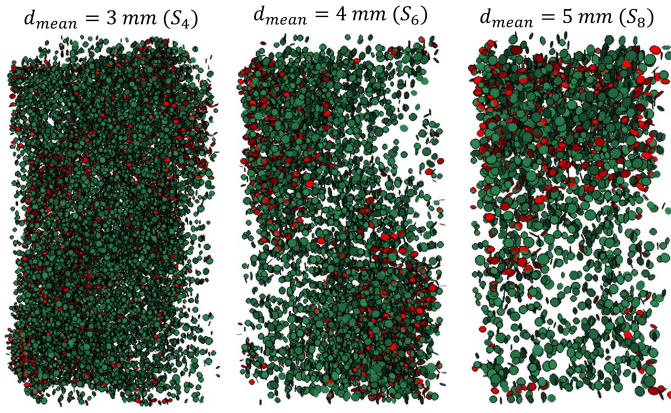


Fig. 6: Microcracks developed post failure of the rock specimen. Green color cracks indicate the tensile cracks and the red color cracks indicate the shear cracks.

proposed by (David et al., 1994), can be expressed as:

$$k_{\sigma} = k_i e^{-\gamma(\sigma_e - \sigma_i)} \quad (5)$$

Here, k_{σ} represents the permeability at effective stresses σ_e , while k_i is the permeability at the initial effective stress σ_i . The stress-sensitive coefficient, γ , is a parameter that determines how sensitive the rock is to the applied stress. A higher value of γ implies that the rock is more sensitive to stress and its permeability reduces steeply. David et al. (1994) suggested that γ is the product of the pore-sensitive coefficient and rock compressibility. Chen et al. (2016) extended the exponential model to fractured porous rock by formulating a semi-empirical model in which γ is a non-linear function of effective stress and pore compressibility.

The other popular model, the power law model, can be expressed as (Shi and Wang, 1986):

$$k_{\sigma} = k_i \sigma_e^{-b} \quad (6)$$

Here, b is a material constant, and k_{σ} and k_i have the same meanings as in the exponential model. Chen et al. (2016) demonstrated that the exponential form of the stress-sensitive model fits data from various fields, but there are cases where the power law model is more appropriate (Nolte et al., 2021). There is no solid explanation for the choice of these empirical equations for a particular rock type. We fit the exponential model and the power law model to the simulation data and found that exponential data fits better.

The stress-sensitive coefficient, γ , is calculated for each data set and presented in figure 10. From the figure, it can be observed that the stress-sensitive coefficient is changing from $1.060 \times 10^{-3} MPa^{-1}$ to $1.085 \times 10^{-3} MPa^{-1}$.

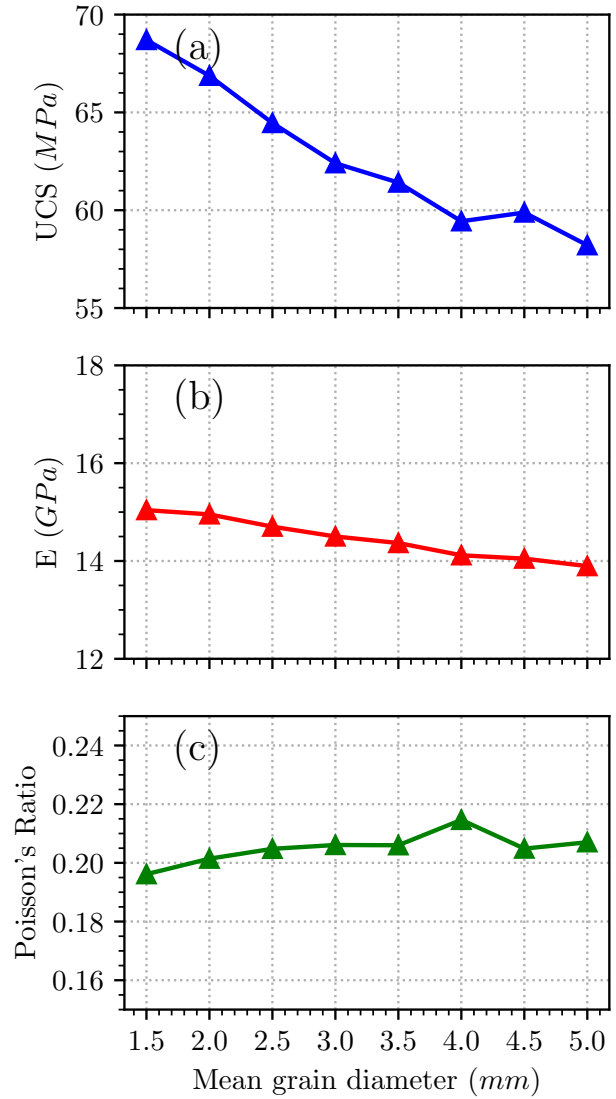


Fig. 7: Variation of mechanical properties with different grain sizes, (a) UCS (b) Young's Modulus (c) Poisson's ratio.,

Chen et al. (2016) compiled the stress-sensitive coefficients for different porous and fractured rock types. They used a variable stress-sensitivity coefficient in their study. The stress-sensitive coefficient obtained in the present study is in the same order as the compiled values of Chen et al. (2016).

The variation of γ with the grain size is shown in figure 10. It can be seen from the figure that a decreasing trend in γ with an increase in grain size. However, the variation in γ is very small compared to the experimental data reported by other researchers (Chen et al., 2016). Generally, sandstone and limestone porous rocks exhibit a variation in γ in the order of $1 - 10 \times 10^{-3} MPa^{-1}$, whereas in the present study, the variation is in the order of $0.01 \times 10^{-3} MPa^{-1}$ (figure 10). It is widely recognized that when a solid un-

Table 3: PFC3D simulation results

Sample	Mean grain size (mm)	UCS (MPa)	Young's modulus (GPa)	Poisson's ratio	Initial permeability (m^2)	$\gamma(10^{-3}MPa^{-1})$
S1	1.5	68.7	15.0	0.196	7.671947e-10	1.0836
S2	2.0	66.9	15.0	0.201	1.370180e-09	1.0838
S3	2.5	64.5	14.7	0.205	2.143630e-09	1.0810
S4	3.0	62.4	14.5	0.206	3.134392e-09	1.0767
S5	3.5	61.4	14.4	0.206	4.319587e-09	1.0705
S6	4.0	59.4	14.1	0.215	5.723061e-09	1.0704
S7	4.5	59.9	14.0	0.205	7.260408e-09	1.0607
S8	5.0	58.2	13.9	0.207	9.223588e-09	1.0606
Variance	1.5	1.39e+01	1.85e-01	2.89e-05	8.868306e-18	8.9055e-05

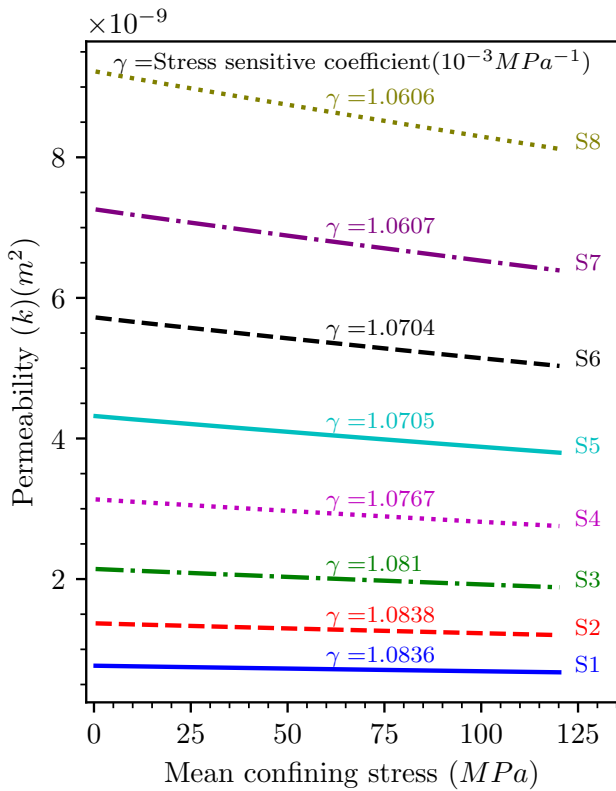


Fig. 8: Permeability variation with increasing confining pressure for samples S1 to S8 increasing grain size

dergoes elastic deformation and contains a singular pore channel, the sensitivity to pressure is significantly greater for pores that resemble cracks as opposed to those with equant shapes. Therefore, it is anticipated that any alterations in permeability as a result of changes in pressure will be dictated by the distribution of pore shapes rather than by general parameters such as porosity (David et al., 1994). However, in our specific study, the grains of the material under investigation are well-sorted, resulting in equant or regular pore spaces. These types of pore shapes are known to be less sensitive to stress changes compared

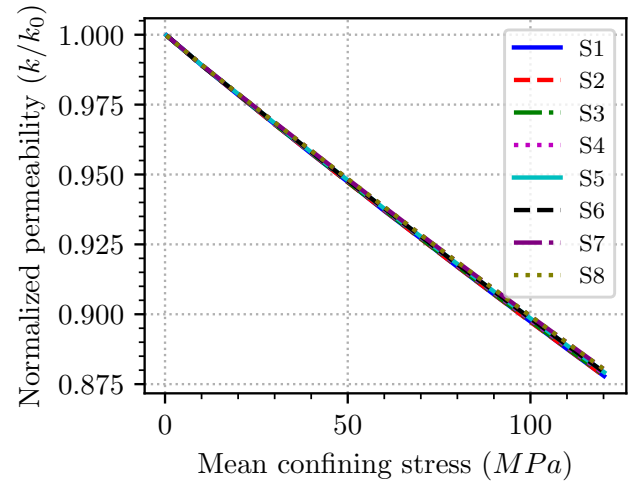


Fig. 9: Normalized permeability variation with the applied mean confining stress. It can be seen that all the lines are overlapped because the variation is very little.

to micro-cracks or elliptical pores. Therefore, our research suggests that the sensitivity of the material to changes in pressure may be different than what has been previously observed in other materials with different pore distributions.

The findings of the present study that indicate higher γ for finer pores align with previous research (Dong et al., 2010). However, it would be premature to conclude that γ decreases with increasing grain size due to several factors. Firstly, the observed variance in γ ($0.000089 \times 10^{-3}MPa^{-1}$) is negligible (table 3). Secondly, the initial permeabilities of different samples are not the same (table 3). Thirdly, the standard deviation of the grain size distribution is not the same (figure 2).

Larger mineral grains result in more stress concentration under the same loading conditions, and specimens with bigger grain sizes fail at lower loading levels, ultimately affecting the UCS of rocks (Yu et al., 2018). While it may

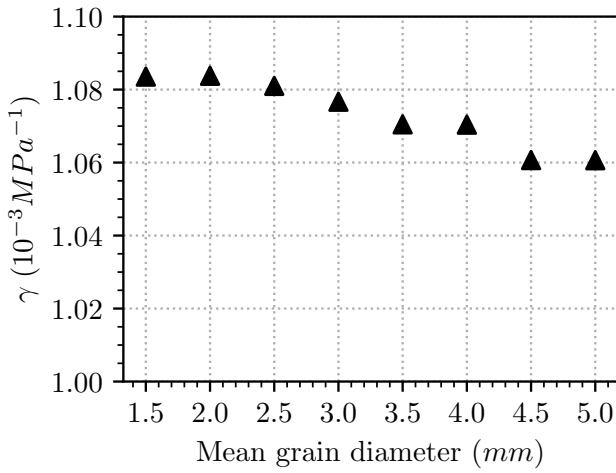


Fig. 10: Variation of stress-sensitive coefficient (γ) with different grain sizes

be true for the UCS, it may not be directly applicable to the pore volume deformation and permeability. From figure 11, it can be seen that the same variation observed in the mechanical properties is not seen in the variation of γ value. Furthermore, Gangi (1978) modeled the variation of permeability with the stress of intact rocks using the concept of deformable grains and Hertz theory. They argued that as long as the standard deviation of the log-normal distribution of grain sizes is small, the mean grain size itself is unlikely to have a significant impact on the permeability change. However, the grains in the present study are rigid, and the pore volume deformation is a result of grain displacements and rearrangement rather than the grains themselves changing shape. In this instance also, it is evident that the variation in grain size has very little impact on the stress-dependent permeability, as illustrated in figure 9. Furthermore, granular rocks that exhibit porosity, such as sandstones, are characterized by grains that lack a close-fitting arrangement, with their cohesion dependent upon a binding agent. This results in the presence of significant intergranular pore spaces (Atapour and Mortazavi, 2018). This suggests that to better understand the permeability phenomena under stress, it is important to examine pore morphology, pore size, and binding material along with the grain size distribution.

Based on our initial research, it appears that further investigation is necessary to fully understand the impact of grain size on our results. To achieve this, we must maintain a fixed initial porosity and permeability while controlling for a constant ratio of maximum to minimum grain sizes and standard deviation. However, achieving these conditions may prove challenging. To gain a more comprehensive understanding of the influence of grain size, it

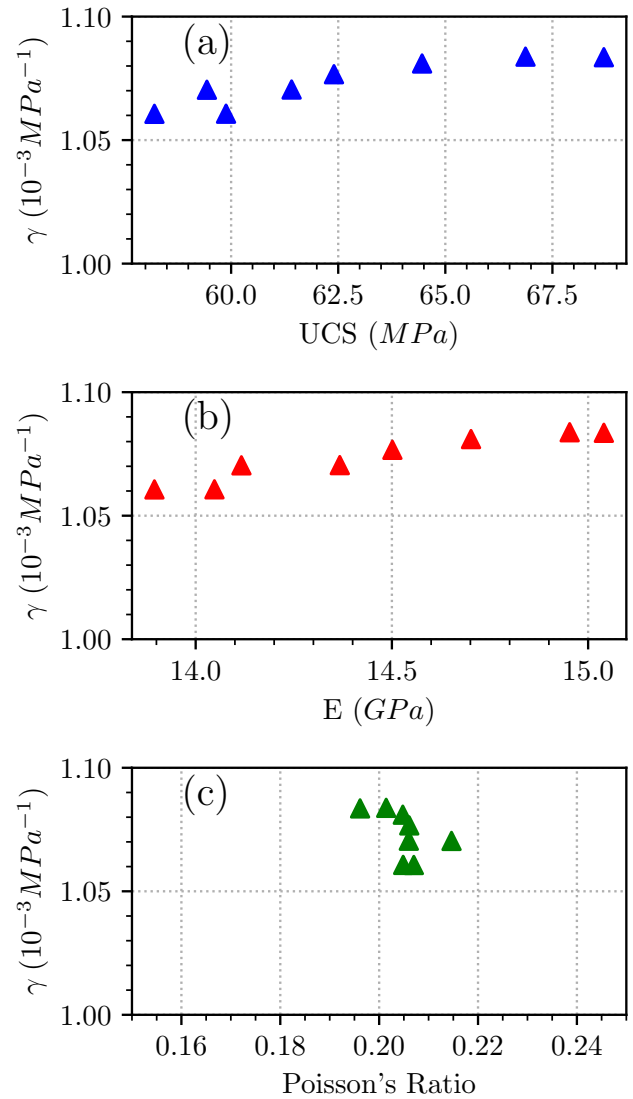


Fig. 11: Variation of stress-sensitive coefficient (γ) with (a) UCS (b) Young's modulus (E) and (c) Poisson's ratio

would be beneficial to incorporate a wider range of grain sizes in our analysis. Unfortunately, the current study is limited by the use of the minimum grain size due to computational constraints. Despite these limitations, we believe that our preliminary findings warrant further investigation and that future research could shed more light on this important area of study.

4 CONCLUSIONS

- In the present study, we applied the discrete element method to generate digital rock samples of varying grain sizes. These digital rocks were calibrated and assigned micro-mechanical properties to mimic moderate-strength porous sandstone rocks. Elastic and strength parameters were obtained by simulat-

ing UCS (uniaxial compressive strength) tests on the digital rocks.

- Our observations indicate that UCS and Young's modulus decrease with increasing grain sizes. However, Poisson's ratio does not follow any correlation with the grain size.
- Our simulation results show that within the studied range, the stress-sensitivity coefficient variation is very small and any definitive conclusions on the grain size-dependent stress-sensitive permeability behavior can not be made. Although the variation is very small, the results indicate that fine-grained rocks display higher stress sensitivity than coarse-grained rocks, likely due to their small pore sizes.
- It is important to note that the stress-sensitivity coefficient variation cannot be solely attributed to the grain size. Other factors, such as initial permeability and porosity, must also be taken into account.
- Our study focused solely on the influence of grain size and did not consider the effects of pore size, pore morphology, cementation, or the presence of micro-cracks. Future research needs to evaluate the effects of these parameters to model the stress-sensitive fluid flow behavior in porous rocks accurately.

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